CSA05

DATABASE MANAGEMENT SYSTEM

Assessment Tool 1 – Short Answer

Q1. UPS Product Tracking Information System

Entities and Attributes

1. Shipped_Item

- Item_Number (Primary Key)
- Weight
- Dimensions
- Insurance_Amount
- Destination
- Final_Delivery_Date

2. Retail_Center

- Unique_ID (Primary Key)
- Type
- Address

3. Transportation_Event

- Schedule_Number (Primary Key)
- Type (Flight/Truck)
- Delivery_Route

Relationships

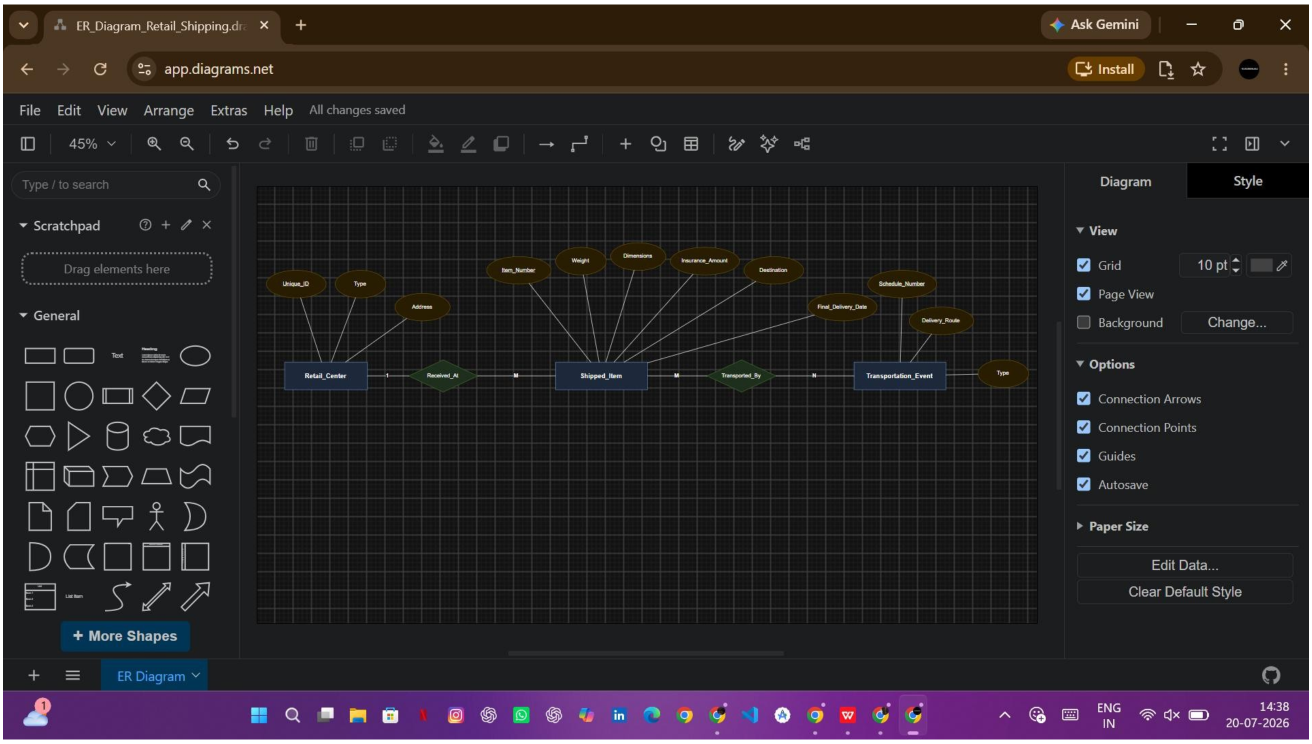
Received_At

- A **Retail Center** receives many **Shipped Items**.
- Each **Shipped Item** is received at only one **Retail Center**.
- **Cardinality: 1 : M**

Transported_By

- A **Shipped Item** may pass through one or more **Transportation Events**.
- One **Transportation Event** can transport many **Shipped Items**.
- **Cardinality: M : N**

ER Diagram



Conclusion

The ER diagram represents the UPS Product Tracking System by identifying the three main entities—**Shipped Item**, **Retail Center**, and **Transportation Event**—along with their relationships and cardinalities.

Q2. Course Management System

Entities and Attributes

1. Course

- Course_Name (Primary Key)
- Department
- Course_Number

2. Section (Weak Entity)

- Section_Number (Partial Key)
- Enrollment

3. Room

- Room_Number (Primary Key)
- Capacity
- Building

4. Exam

- Exam_ID (Primary Key)
- Time

Relationships

Course – Section

- One course has many sections.
- One section belongs to one course.
- **Cardinality: 1 : M**

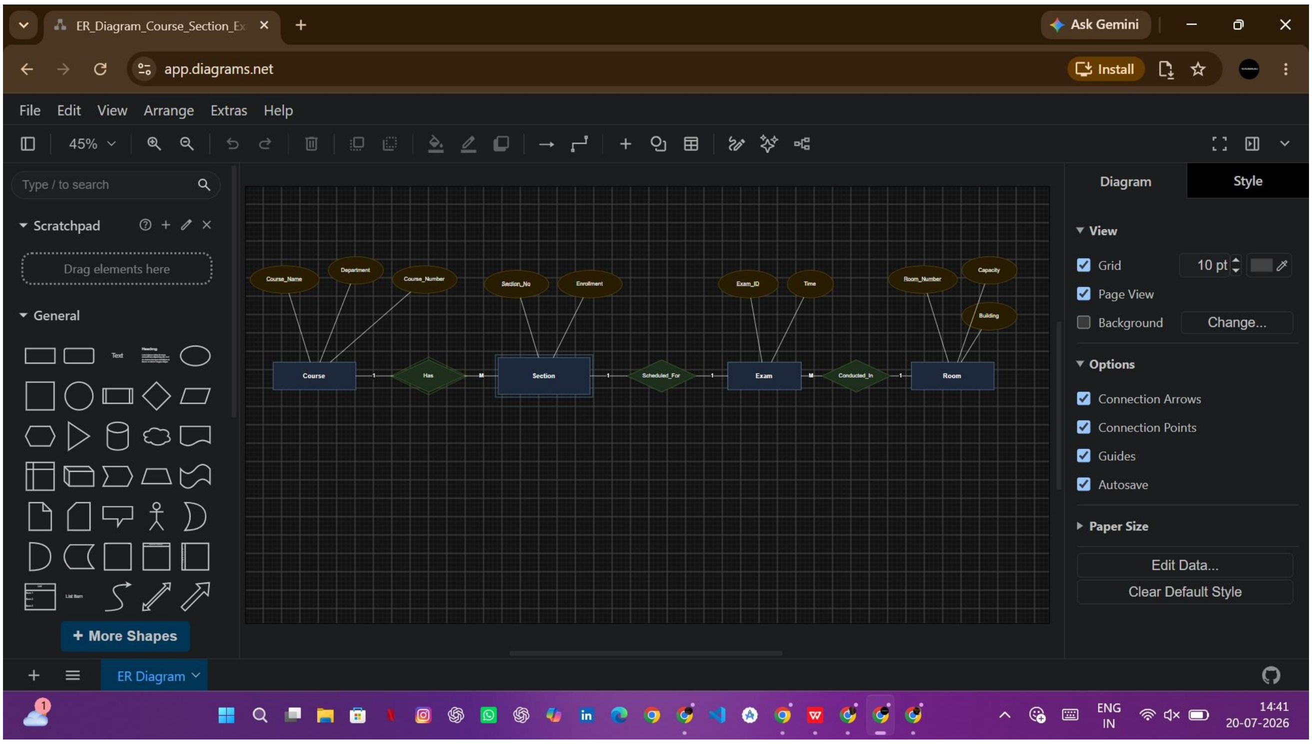
Section – Exam

- Each section has one final exam.
- **Cardinality: 1 : 1**

Exam – Room

- One room may conduct many exams.
- One exam is conducted in one room.
- **Cardinality: M : 1**

ER Diagram



Conclusion

The ER diagram models the examination scheduling system by separating **Course**, **Section**, **Exam**, and **Room** into independent entities, reducing redundancy and improving data organization.

Q3. University Registrar's Office Management System

Entities and Attributes

1. Course

- Course_Number (Primary Key)
- Title
- Credits
- Syllabus
- Prerequisites

2. Course_Offering

- Offering_ID (Primary Key)
- Year
- Semester
- Section_Number
- Timings
- Classroom

3. Student

- Student_ID (Primary Key)
- Name
- Program

4. Instructor

- Instructor_ID (Primary Key)
- Name
- Department
- Title

5. Enrollment (Relationship Entity)

- Grade

Relationships

Course – Course Offering

- One course can have many offerings.
- **Cardinality: 1 : M**

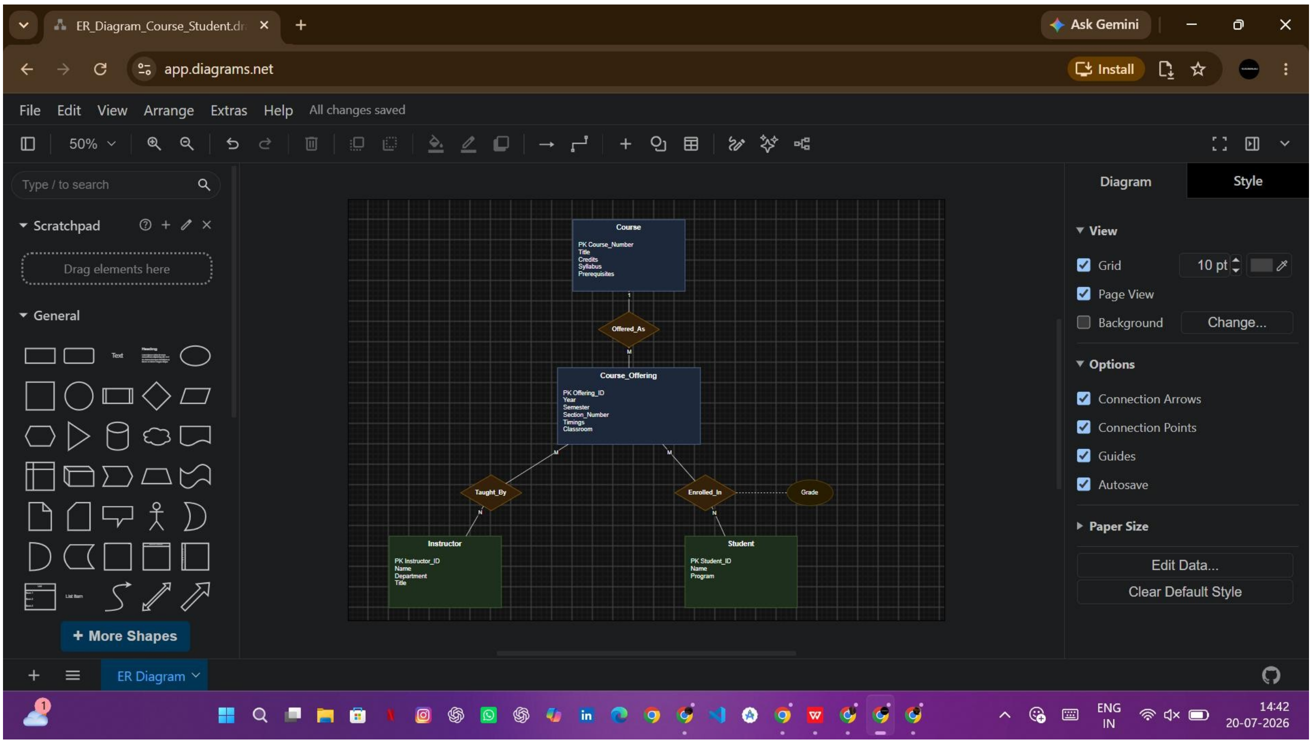
Instructor – Course Offering

- One instructor can teach many course offerings.
- A course offering may have one or more instructors.
- **Cardinality: M : N**

Student – Course Offering

- A student can enroll in many course offerings.
- Each course offering can have many students.
- Grade is stored in the **Enrollment** relationship.
- **Cardinality: M : N**

ER Diagram



Conclusion

The ER diagram effectively models the University Registrar's Office by representing **Course**, **Course Offering**, **Student**, and **Instructor** entities along with their relationships. The **Enrollment** relationship stores the grade obtained by each student in a course offering.

Q4. Hospital Management System

Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram.

1. Introduction

A **Hospital Management System** is used to maintain information about patients, physicians, medical tests, and treatment history. The ER diagram models the relationships between these entities for efficient data management.

2. Entities and Attributes

Entity 1: Patient

Primary Key: SS#

Attributes:

- SS# (Primary Key)
- Name
- Insurance

Entity 2: Physician

Primary Key: Physician_ID *(added for better database design)*

Attributes:

- Physician_ID (Primary Key)
- Name
- Specialization

  Note: Although the question lists only the physician's name, using a unique **Physician_ID** is considered good database design practice.

Entity 3: Test_Log

Primary Key: Test_Log_ID *(or composite key: SS# + Test_Name + Date + Time)*

Attributes:

- Test_Log_ID (Primary Key)
- SS# (Foreign Key)
- Test_Name
- Date
- Time

3. Relationships

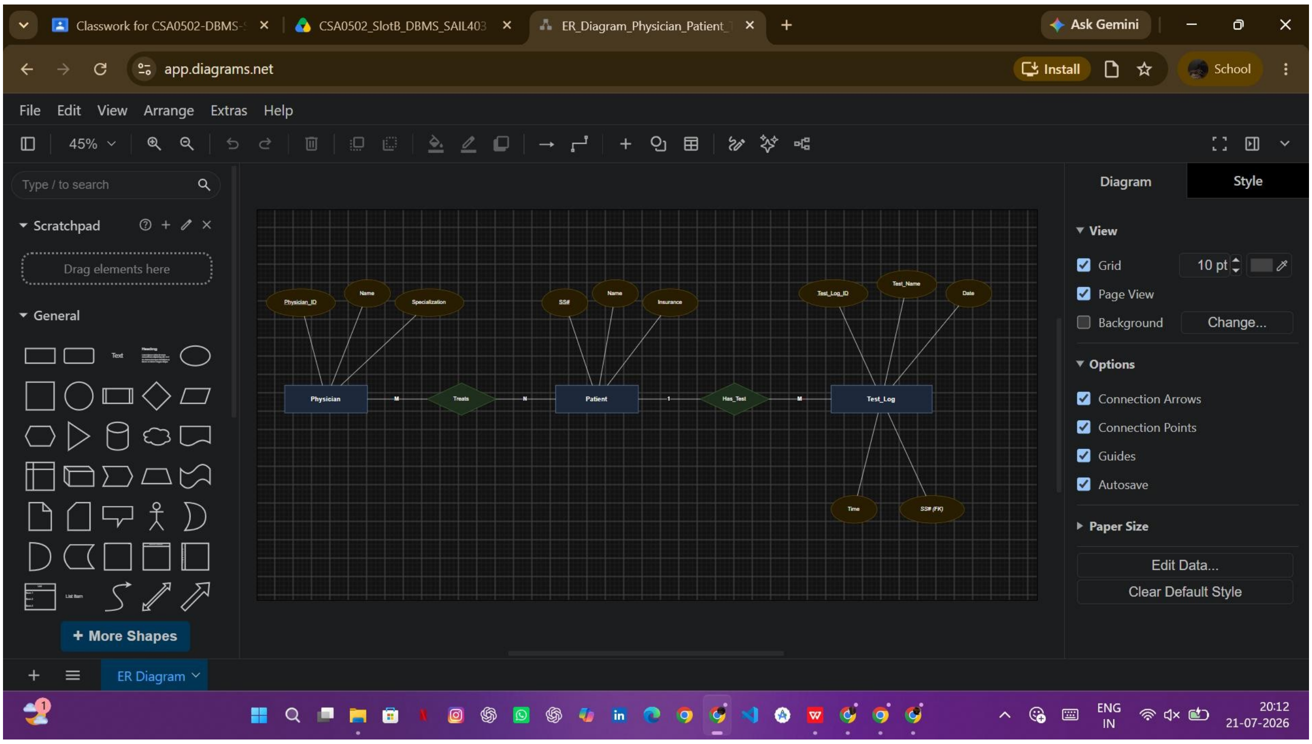
Treats

- A **Physician** can treat many **Patients**.
- A **Patient** may be treated by one or more **Physicians**.
- **Cardinality: M : N**

Has_Test

- One **Patient** can undergo multiple tests.
- Each **Test Log** belongs to one patient.
- **Cardinality: 1 : M**

4. ER Diagram



5. Relational Tables

Table 1: Patient

SS# (PK)	Name	Insurance
P101	Rahul	Star Health
P102	Priya	LIC Health

Table 2: Physician

Physician_ID (PK)	Name	Specialization
D101	Dr. Kumar	Cardiologist
D102	Dr. Meena	Neurologist

Table 3: Test_Log

Test_Log_ID (PK)	SS# (FK)	Test_Name	Date	Time
T001	P101	Blood Test	10-07-2026	09:30 AM
T002	P101	ECG	10-07-2026	11:00 AM
T003	P102	MRI Scan	11-07-2026	02:00 PM

Table 4: Doctor_Patient

Physician_ID (FK)	SS# (FK)
D101	P101
D102	P102

This table resolves the **many-to-many (M:N)** relationship between physicians and patients.

Table 5: Patient_History

SS# (FK)	Test_Name	Date
P101	Blood Test	10-07-2026
P101	ECG	10-07-2026
P102	MRI Scan	11-07-2026

6. Cardinality Summary

Relationship	Cardinality

Relationship	Cardinality
Patient → Test_Log	**1 : M**
Physician ↔ Patient	**M : N**

7. Conclusion

The Hospital Management System ER model consists of three main entities—**Patient**, **Physician**, and **Test_Log**. The **Patient–Physician** relationship is **many-to-many**, while the **Patient–Test_Log** relationship is **one-to-many**. The relational tables efficiently store patient details, physician information, medical tests, and treatment history, ensuring accurate and organized hospital data management.

Q5. Company Project Management System

Construct an ER Diagram for a Company with Departments, Employees, Projects, Dependents, and their relationships.

1. Introduction

A **Company Project Management System** is used to manage information about departments, employees, projects, managers, dependents, and employee project assignments. The ER model helps organize company data efficiently while reducing redundancy.

2. Entities and Attributes

Entity 1: Department

Primary Key: Department_ID

Attributes:

- Department_ID (Primary Key)
- Department_Name (Unique)
- Location

Entity 2: Employee

Primary Key: SSN

Attributes:

- SSN (Primary Key)
- Name
- Address
- Salary
- Sex
- Birth_Date

Entity 3: Project

Primary Key: Project_Number

Attributes:

- Project_Number (Primary Key)
- Project_Name (Unique)
- Location

Entity 4: Dependent (Weak Entity)

Partial Key: Dependent_Name

Attributes:

- Dependent_Name (Partial Key)
- Birth_Date
- Relationship

Dependent is a **Weak Entity** because it cannot exist without an Employee.

3. Relationships

1. Manages

- One employee manages one department.
- Every department has one manager.
- Manager's **Start_Date** is stored in this relationship.
- **Cardinality: 1 : 1**

Relationship Attribute:

- Start_Date

2. Works_For

- One department has many employees.
- Every employee belongs to only one department.
- **Cardinality: 1 : M**

3. Controls

- One department controls many projects.
- Every project belongs to one department.
- **Cardinality: 1 : M**

4. Works_On

- One employee can work on many projects.
- One project can have many employees.
- **Cardinality: M : N**

Relationship Attribute:

- Hours_Per_Week

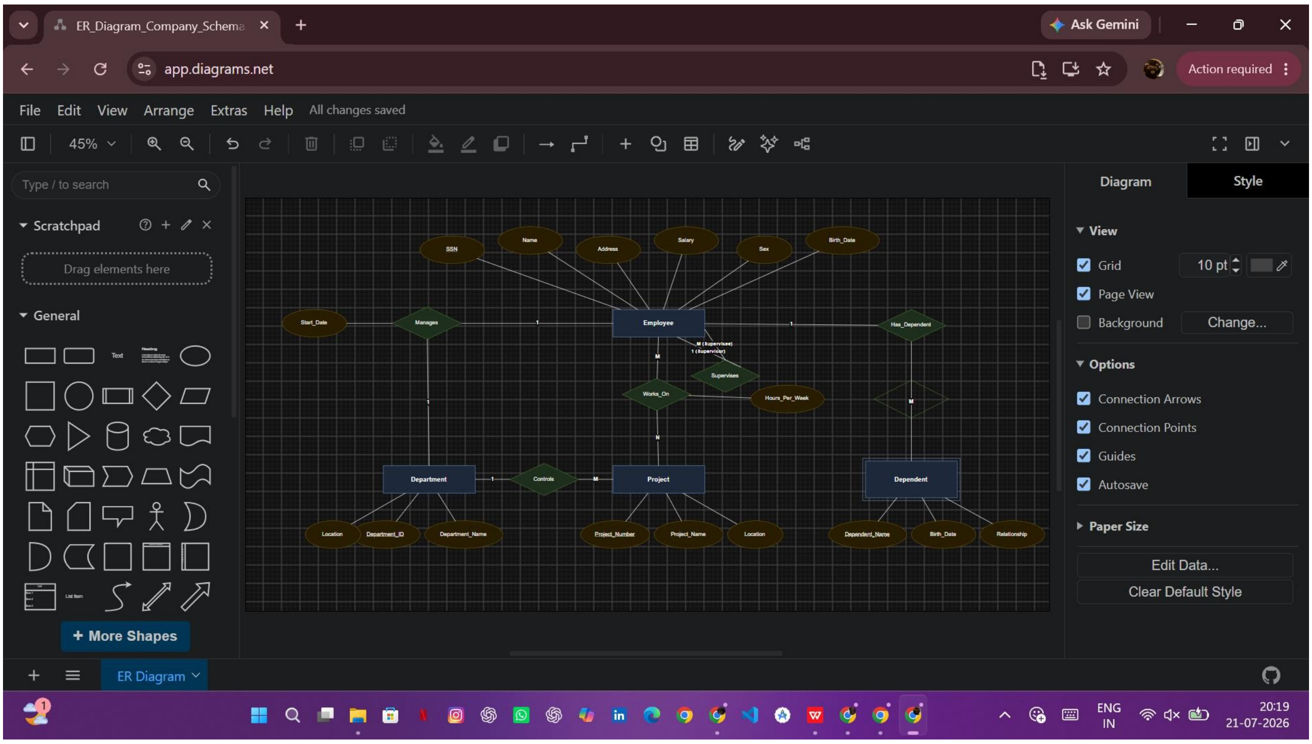
5. Supervises

- One employee supervises many employees.
- Each employee has only one immediate supervisor.
- **Cardinality: 1 : M** (Recursive Relationship)

6. Has_Dependent

- One employee can have many dependents.
- Each dependent belongs to one employee.
- **Cardinality: 1 : M**

4. Complete ER Diagram



5. Cardinality Summary

Relationship	Cardinality
Department → Employee	**1 : M**
Department → Project	**1 : M**
Employee ↔ Project	**M : N**
Employee → Dependent	**1 : M**
Employee → Employee (Supervisor)	**1 : M**
Employee ↔ Department (Manager)	**1 : 1**

6. Relational Schema

Department

Department(Department_
ID (PK),
Department_Name,
Location
)

Employee

Employee(SS
N (PK),
Name,
Address,
Salary,
Sex,
Birth_Date,
Department_ID (FK),
Supervisor_SSN (FK)
)

Project

Project(
Project_Number (PK),
Project_Name,
Location,
Department_ID (FK)
)

Works_On

Works_On(S
SN (FK),
Project_Number (FK),
Hours_Per_Week
)

Manages

Manages(SS
N (FK),
Department_ID (FK),
Start_Date
)

Dependent

Dependent(

SSN (FK),
Dependent_Name,
Birth_Date,
Relationship
)

7. Conclusion

The Company Project Management System ER model consists of four main entities: **Department**, **Employee**, **Project**, and **Dependent**. It includes key relationships such as **Manages**, **Works_For**, **Controls**, **Works_On**, **Supervises**, and **Has_Dependent**. The model accurately represents project assignments, department management, employee supervision, and dependent information while maintaining data integrity and minimizing redundancy. This design is suitable for efficient database implementation in a real-world company management system.

Q6. Explain Schema and Instance with Examples. Distinguish between Logical Schema, Physical Schema, and View Schema.

1. Introduction

A **Database Management System (DBMS)** organizes data using different levels of schemas. A **schema** defines the structure of the database, while an **instance** represents the actual data stored in the database at a particular point in time.

2. Schema

Definition

A **schema** is the overall logical design or blueprint of a database. It specifies the database structure, including tables, attributes, relationships, constraints, and data types.

Characteristics

- Defines the database structure.
- Created during database design.
- Changes rarely.
- Also known as **database blueprint**.

Example

Consider a **Student** table:

```
Student(
   Student_ID INT PRIMARY KEY,
   Name VARCHAR(50),
   Department VARCHAR(30),
   CGPA DECIMAL(3,2)
)
```

The above table definition represents the **schema** because it specifies the structure of the database.

3. Instance

Definition

An **instance** is the actual collection of data stored in the database at a specific point in time.

Characteristics

- Represents the current contents of the database.
- Changes frequently when records are inserted, updated, or deleted.
- Also known as the **database state**.

Example

Student_ID	Name	Department	CGPA
101	Rahul	CSE	8.75
102	Priya	AI & ML	9.10
103	Arjun	IT	8.45

The above records represent one **instance** of the database.

4. Difference Between Schema and Instance

Schema	Instance
Defines the database structure.	Represents the actual data stored.
Changes rarely.	Changes frequently.
Created during database design.	Updated whenever data changes.
Also called the database blueprint.	Also called the database state.

5. Types of Schema

DBMS follows the **Three-Schema Architecture**, which consists of:

1. Logical Schema

2. Physical Schema
3. View (External) Schema

A. Logical Schema

Definition

The **Logical Schema** describes the logical structure of the entire database, including tables, attributes, relationships, and constraints, without specifying how the data is physically stored.

Characteristics

- Represents the conceptual design of the database.
- Independent of storage details.
- Used by database designers and developers.

Example

```
Employee( Em
   p_ID, Name,
   Salary,
   Department
)
```

B. Physical Schema

Definition

The **Physical Schema** describes how the data is actually stored in the storage devices, including file organization, indexing, and storage allocation.

Characteristics

- Defines physical storage details.
- Includes indexes and storage methods.
- Managed by the DBMS.
- Used by database administrators (DBAs).

Example

- Data stored in disk blocks.
- B+ Tree indexes.
- Hash files.
- File organization techniques.

C. View Schema (External Schema)

Definition

A **View Schema** defines how different users see the database. It displays only the required portion of the database and hides unnecessary or sensitive information.

Characteristics

- Provides customized views for users.
- Enhances data security.
- Simplifies database access.
- Multiple view schemas can exist for one database.

Example

Student Table

Student_ID	Name	Department	CGPA
101	Rahul	CSE	8.75
102	Priya	AI & ML	9.10

Faculty View

Student_ID	Name	CGPA
101	Rahul	8.75
102	Priya	9.10

Only the required information is displayed.

6. Difference Between Logical, Physical, and View Schema

Logical Schema	Physical Schema	View Schema
Describes the logical structure of the database.	Describes how data is physically stored.	Describes how users view the database.
Independent of storage.	Depends on storage devices.	Independent of physical storage.
Used by database designers.	Used by DBAs.	Used by end users and applications.
Defines tables, attributes, and relationships.	Defines files, indexes, and storage methods.	Displays only required data.
One logical schema exists.	One physical schema exists.	Multiple view schemas can exist.

7. Three-Schema Architecture Diagram

```
   Users / Applications
           |
+----------------------+
|   View Schema        |
| (External Level)     |
+----------------------+
           |
+----------------------+
|  Logical Schema      |
| (Conceptual Level)   |
+----------------------+
           |
+----------------------+
| Physical Schema      |
| (Internal Level)     |
+----------------------+
           |
    Physical Database
```

8. Conclusion

A **schema** defines the structure of a database, whereas an **instance** represents the data stored at a particular moment. The **Logical Schema** describes the overall database structure, the **Physical Schema** explains how data is stored internally, and the **View Schema** provides customized views for different users. Together, these three schemas form the **Three-Schema Architecture**, ensuring data abstraction, security, and efficient database management.